State Changes

Because there are only two conditions possible when reading or writing digital inputs and outputs—high and low—we can use this predictability to detect a **state change**, where a pin changes from high to low or low to high, or even to count these changes. To detect a state change on a digital input, we don't actually need to know the precise state that the pin is in. Rather, we only need to know when a pin has changed from one state to another.

To do this, we need a way to compare the pin's current state with the state of the pin the last time we read it. If we check an input and it's high but the last time we checked it was low, then the button has been pressed. On the other hand, if we check an input and it's low but last time we checked it was high, then the button is no longer being pressed. By looking for this change, what we are doing is called **edge detection** because really all we are looking for is that specific moment or edge when the state changes from one condition to another.

Let's back up for a second and take a look at a very simple pushbutton circuit, as shown in Figures 5-6 and 5-7. This circuit will use the built-in LED on pin 13 (not shown) and add a momentary pushbutton on pin 2, which only stays closed or on when the button is actively being pressed. This will provide a good basis for our further discussion on state changes.

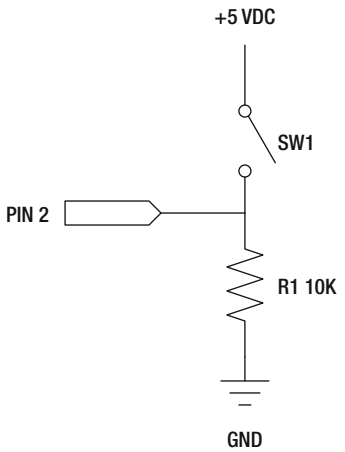

Figure 5-6. *Pushbutton schematic*